

popquiz equilibrium

⚠ This is a preview of the draft version of the quiz

Started: Apr 29 at 7:57pm

Quiz Instructions



Question 1 25 pts



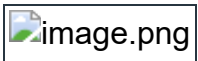
hint: see picture below. The magnitude of the tension is the same in both of the strings. $T_1 = T_2 = T$

1) The Sum(up) = sum(down) so $T_y + T_y = \text{weight}$.

solve for T_y . then

2) Use your trig skills to find T using : $T_y = T \times \dots(40)$ is sine? or cosine? or tan ?

solve for T .



389N



778N



500N



326N



Question 2 25 pts



What is true?



$T_1 = T_2 = 250\text{N}$



$T_2 = 250\text{N}, T_1 = 0$



$T_1 = T_2 = 500\text{N}$



$T_2 = 500\text{N}$, $T_1 = 0\text{N}$



Question 3 25 pts



What is the normal force?

hint: the pull P has 2 components shown below. P_x and P_y .

$\text{sum}(\text{up}) = \text{sum}(\text{down})$

$\text{Equ1 } N + P_y = 400$

Use your trig skill to find $P_y = P \times \dots (30)$ with $P = 100\text{N}$. Is ... cosine ? sine? tangent ?

plug P_y in equation1 and solve for N .



100N



350N



400N



313N



Question 4 25 pts



The crate at the origin is in equilibrium and is pulled by 3 strings. The 2 blue strings make an angle of 30 degrees . The tensions in the blue string are T_1 and T_2 . $T_1 = T_2 = T$ in magnitude = 10N.

What is the tension in the orange string? (T_3)

hint: $\text{sum}(\text{right}) = \text{sum}(\text{left})$ so $T_3 = T_x + T_x$ use your trig skills

☐

20N

☐

5N

☐

17.3N

☐

10N

Not saved

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